

Clustering

AIM:

To group customers using K means clustering & find the optimal number of clustering with the elbow method.

ALGORITHM:

- i) Import necessary libraries
- ii) Load Dataset & select features
- iii) Apply K-Means clustering
- iv) Use Elbow method to determine the best k
- v) Visualize clusters with scatter plot

PROGRAM:

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler
import seaborn as sns

df = pd.read_csv("Mall customers.csv")
```

KMeans = KMeans (n_clusters = 5, random_state = 0)

df['cluster'] = KMeans . fit_predict()

distortions = []

for i in range(1, 11):

km = KMeans (n_clusters = i)

km = fit(df['Annual Income (k\$)', 'Spending
score (1-100)'])

distortions . append (km.inertia)

plt . plot (range(1, 11), distortions, marker = "o")

plt . title ("Elbow Method")

plt . ylabel ("Inertia")

plt . xlabel ("No. of clusters")

plt . show()

sns . scatterplot (data = df, x = "Annual Income (k\$)",

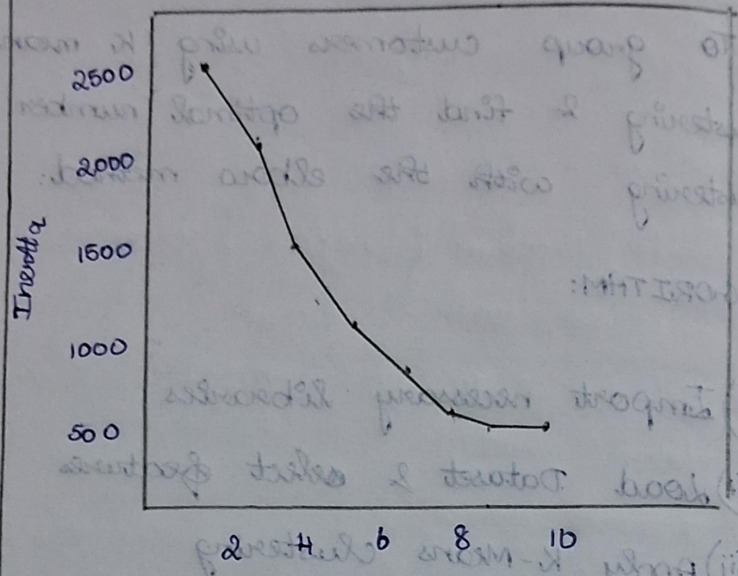
y = "Spending score (1-100)", hue = "cluster", palette = "set1")



RESULT:

Thus the customers are successfully grouped into distinct clusters showing spending habit pattern

Elbow Method



Annual Income (K \$)

